BIN DERANGEMENTS AND THE GATE WIDTH THEOREM

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Abstract. For any prime $p > b$, define the collision count $C(g) = \#\{r : \lfloor br/p \rfloor = \lfloor b(gr \bmod p)/p \rfloor\}$. We prove that the set of multipliers g with $C(g) = 0$ (bin derangements) is exactly $\{-u/(b-u) \bmod p : u = 1, \ldots, b-1\}$. In particular, there are exactly $b-1$ such elements, independent of the prime. No primitive-root hypothesis is needed. The key observation is that the map $r \mapsto br \bmod p$ sends contiguous digit bins to residue classes modulo b.

When b is a primitive root modulo p, the bin derangements coincide with the zero lags of the cyclic autocorrelation.

For the non-deranging multipliers $g \neq 1$ (those with $C(g) > 0$), the mean collision count satisfies an exact formula:

$$\overline{C} = \frac{Q\big(b(Q-1) + 2R\big)}{b(Q-1) + R},$$

where $Q = \lfloor (p-1)/b \rfloor$ and $R = (p-1) \bmod b$. For the residue class $R = 0$, this simplifies to $\overline{C} = Q$ exactly. For $Q = 1$ (equivalently, $b + 1 < p \leq 2b + 1$), the mean is always 2.

Together, these results decompose the autocorrelation into two independent controls: the base determines how many lags produce zero autocorrelation ($b - 1$), while the prime determines which lags are blocked and how strongly the surviving lags contribute.

1. Introduction

This paper characterizes the zero set of the autocorrelation.

The digit function $\delta(r) = \lfloor br/p \rfloor$, which assigns to each residue $r \bmod p$ its leading digit in base b, has been studied in connection with the distribution of digits of primes [2], exponential sums over digit constraints [4], and classical properties of positional expansions [1]. The function partitions $\{1, \ldots, p-1\}$ into b bins $B_0, \ldots, B_{b-1}$, where

$$B_d = \{r \in \{1, \ldots, p-1\} : \delta(r) = d\}.$$

These bins form a partition by contiguous intervals: the bin sizes are $\lfloor (d+1)p/b \rfloor - \lfloor dp/b \rfloor$, determined directly by interval arithmetic.

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When b is a primitive root modulo p, the autocorrelation of the repetend of $1/p$ at lag ℓ counts positions j where the digit at position j equals the digit at position $j+\ell$:

$$R(\ell) = \left|\{j \in \{0,\dots,L{-}1\} : \delta(b^j) = \delta(b^{j+\ell})\}\right| = \sum_{d=0}^{b-1} |B_d \cap gB_d|,$$

where $g = b^\ell \bmod p$ and the intersection is under multiplication in $(\mathbb{Z}/p\mathbb{Z})^\times$.

Without the primitive-root hypothesis, the same formula defines the collision count

$$C(g) = \sum_{d=0}^{b-1} |B_d \cap gB_d|$$

for an arbitrary multiplier $g \in \mathbb{F}_p^\times$, and the algebraic arguments below work directly with $C(g)$.

The question of when $R(\ell) = 0$ reduces to a combinatorial question about the Beatty partition: for which group elements g does multiplication by g move every residue to a different bin? Such elements are *derangements* of the bin partition, in the sense of the classical theory of permutation derangements [6]. We determine their exact number and give an explicit formula.

2. The Bin Partition

The bins $B_0, \dots, B_{b-1}$ are contiguous intervals of $\{1,\dots,p{-}1\}$ determined by the floor function $\delta(r) = \lfloor br/p \rfloor$. Their sizes are controlled by the division $p{-}1 = bQ{+}R$, where $Q = \lfloor (p{-}1)/b \rfloor$ and $R = (p{-}1) \bmod b$: exactly R bins have size $Q+1$ and $b-R$ bins have size Q. Equivalently,

$$|B_d| = \left\lfloor \frac{(d+1)p}{b} \right\rfloor - \left\lfloor \frac{dp}{b} \right\rfloor.$$

When $p \le b+1$, each bin contains at most one residue, the partition is injective, and the autocorrelation is trivially zero at all non-zero lags. When $p > b+1$, multiple residues share bins, and non-trivial autocorrelation structure emerges.

3. Bin Derangements

Definition 1. An element $g \in (\mathbb{Z}/p\mathbb{Z})^*$ is a *bin-deranging multiplier* (or *bin derangement*) if

$$gB_d \cap B_d = \varnothing \quad \text{for every } d \in \{0,\dots,b{-}1\}.$$

That is, multiplication by g moves every residue out of its digit bin. The terminology follows the classical notion of a derangement (a permutation

with no fixed points) [6], extended to the partition setting: no element remains in its original bin block.

Proposition 2. *For a prime p where b is a primitive root, $R(\ell) = 0$ if and only if $g = b^\ell \bmod p$ is a bin-deranging multiplier.*

Proof. $R(\ell) = \sum_d |B_d \cap gB_d|$ where $g = b^\ell$. Each term is non-negative, so $R(\ell) = 0$ if and only if every term vanishes, which is the bin-derangement condition. □

The bin-derangement condition has a geometric interpretation. The bins $B_0, \ldots, B_{b-1}$ are contiguous intervals in $\{1, \ldots, p-1\}$. Multiplication by g permutes the nonzero residues modulo p. A bin derangement is a multiplier for which no bin meets its image.

4. The Gate Width Theorem

Lemma 3 (Conjugation)**.** *For each $r \in \{1, \ldots, p-1\}$, let $x = [br]_p$ (the least positive residue of br modulo p). Then $br = p\,\delta(r) + x$, and consequently*

$$\delta(r) = \delta(s) \quad \Longleftrightarrow \quad [br]_p \equiv [bs]_p \pmod{b}.$$

Under the permutation $r \mapsto [br]_p$, the digit bins are exactly the residue classes modulo b.

Proof. Since $p \nmid br$, the least positive residue of br modulo p is the remainder after division: $br = p\lfloor br/p \rfloor + [br]_p = p\,\delta(r) + x$. Reducing modulo b: $0 \equiv p\,\delta(r) + x \pmod{b}$. Since $\gcd(p, b) = 1$, the residue class of x modulo b determines $\delta(r)$ uniquely. □

Lemma 4. *For every $g \in \mathbb{F}_p^\times$,*

$$C(g) = \#\{x \in \{1, \ldots, p-1\} : x \equiv [gx]_p \pmod{b}\}.$$

Proof. The map $r \mapsto x = [br]_p$ is a permutation of $\{1, \ldots, p-1\}$. Apply Lemma 3 to the pair r and $gr \bmod p$. □

Lemma 5. *Fix $g \neq 1$ and define $c \equiv b(1-g)^{-1} \pmod{p}$ with $1 \leq c \leq p-1$. If $1 \leq c \leq b-1$, then $C(g) = 0$.*

Proof. The key geometric idea is that when c is small (between 1 and $b-1$), the shifted sequence $mc \bmod p$ stays too close to zero to reach the reference sequence mb, preventing any collision.

A collision gives $y = [gx]_p$ with $y = x + mb$ for some integer m, and $x \equiv -mc \pmod{p}$.

If $m \geq 0$: from $x + mb \leq p-1$, $1 \leq x + mc \leq p-1-m(b-c) < p$, but $x + mc \equiv 0 \pmod{p}$, a contradiction.

If $m < 0$: write $m = -n$ with $n \geq 1$; from $x - nb \geq 1$, $0 < x - nc = (x-nb) + n(b-c) \leq p-1$, again contradicting $x - nc \equiv 0 \pmod{p}$. □

Lemma 6. *If $c \geq b+1$, then $C(g) \geq 1$.*

Proof. Conversely, when c is large enough ($c \geq b+1$), the gap between the shifted and reference sequences is wide enough that a collision point can be constructed explicitly.

Take $x = p - c$. Then $1 \leq x \leq p-b-1$, so $y := x + b = p - c + b$ lies in $\{1, \ldots, p-1\}$ with $y \equiv x \pmod{b}$. Since $x \equiv -c \pmod{p}$ and $c(1-g) \equiv b$, we get $(1-g)x \equiv -b$, hence $gx \equiv x + b = y \pmod{p}$. So x contributes a collision. □

Theorem 7 (Gate width)**.** *Let p be prime and $2 \leq b < p$. Then*

$$\{g \in \mathbb{F}_p^\times : C(g) = 0\} = \left\{ -\frac{u}{b-u} \pmod{p} : u = 1, \ldots, b-1 \right\}.$$

In particular, there are exactly $b-1$ bin-deranging elements. No primitive-root hypothesis is needed.

Proof. By Lemmas 5 and 6, $C(g) = 0$ if and only if $c = b(1-g)^{-1} \in \{1, \ldots, b-1\}$. Solving: $g = 1 - b/c$. Setting $u = b - c$ gives $g = -u/(b-u) \pmod{p}$ for $u = 1, \ldots, b-1$. These are distinct: $-u_1/(b-u_1) \equiv -u_2/(b-u_2) \pmod{p}$ implies $b(u_1 - u_2) \equiv 0 \pmod{p}$, and since $p > b$ this gives $u_1 = u_2$. □

Corollary 8. $\left| \bigcup_{d=0}^{b-1} B_d/B_d \right| = p - b.$

Proof. For $g \in \mathbb{F}_p^\times$, the condition $g \in \bigcup_{d=0}^{b-1} B_d/B_d$ means that $g = xy^{-1}$ for some $x, y \in B_d$, equivalently $gy = x \in B_d$ for some $y \in B_d$. Thus g lies in the union if and only if it is not bin-deranging. By Theorem 7, exactly $b-1$ of the $p-1$ units are deranging, so the union has size $(p-1) - (b-1) = p - b$. □

Remark 9. The key step is the conjugation (Lemma 3): the permutation $r \mapsto [br]_p$ transforms the contiguous Beatty bins into residue classes modulo b. The partition geometry linearizes completely under this change of variables. The squeezing argument in Lemmas 5 and 6 then determines the zero set exactly.

Remark 10. The deranging multipliers $g = -u/(b-u) \pmod{p}$ form a rational family. For $u = b/2$ (when b is even), $g \equiv -1 \pmod{p}$, the complement element. The remaining derangers come in inverse pairs: $-u/(b-u)$ and $-(b-u)/u$ are mutual inverses modulo p.

Remark 11. The number $b-1$ depends only on the base. It does not depend on the prime, the spectral class, or the scale. The gate width is a property of the coordinate system, not of the object being observed. What varies with the prime is *which* $b-1$ lags are blocked, and *how strongly* the surviving lags contribute.

5. The Sum Rule and Mean Formula

The autocorrelation satisfies a simple double-counting identity connecting the collision count to the zeroth bin-size moment.

Proposition 12 (Sum rule)**.**

$$\sum_{g\in\mathbb{F}_p^\times} C(g) = \sum_{d=0}^{b-1} n_d^2,$$

where $n_d = |B_d|$ *is the size of the* d*-th digit bin.*

Proof. $\sum_g C(g) = \sum_g \sum_r [\delta(r) = \delta(gr)] = \sum_r \sum_g [\delta(gr) = \delta(r)]$. For each r, as g ranges over $\mathbb{F}_p^\times$, the element gr ranges over all of $\mathbb{F}_p^\times$. The inner sum counts how many residues share a bin with r, which is $n_{\delta(r)}$. Summing over r: $\sum_r n_{\delta(r)} = \sum_d n_d^2$. □

The bin sizes for a prime $p > b+1$ are determined by the division $p-1 = bQ + R$, where $Q = \lfloor (p-1)/b \rfloor$ and $R = (p-1) \bmod b$. There are R bins of size $Q+1$ and $b-R$ bins of size Q. Hence

$$\sum_{d=0}^{b-1} n_d^2 = R(Q+1)^2 + (b-R)Q^2 = bQ^2 + R(2Q+1).$$

Theorem 13 (Mean constructive value)**.** *For* $p > b+1$ *(so that constructive multipliers exist), the mean of* $C(g)$ *over the* $p-b-1$ *constructive multipliers is*

$$\overline{C} = \frac{Q\big(b(Q-1) + 2R\big)}{b(Q-1) + R}.$$

When b *is a primitive root modulo* p*, this equals the mean of* $R(\ell)$ *over constructive lags.*

Proof. There are $p-2$ multipliers $g \neq 1$, of which $b-1$ have $C(g) = 0$ by Theorem 7. The number of constructive multipliers is $p-2-(b-1) = p-b-1 = b(Q-1)+R$.

The sum over constructive multipliers equals

$$\sum_{g\in\mathbb{F}_p^\times} C(g) - C(1) = bQ^2 + R(2Q+1) - (p-1) = bQ^2 - bQ + 2RQ = Q(b(Q-1)+2R).$$

Dividing: $\overline{C} = Q(b(Q-1) + 2R)/(b(Q-1) + R)$. □

Corollary 14. (1) *For $R = 0$ (class $p \equiv 1 \pmod{b}$): $\overline{C} = Q$ exactly.*
(2) *For $Q = 1$ (equivalently, $b + 1 < p \leq 2b + 1$): $\overline{C} = 2$.*
(3) *As $Q \to \infty$ (large primes in any class): $\overline{C}/Q \to 1$, equivalently $\overline{C} = Q + O(1)$.*

Proof. (1) Set $R = 0$: the formula gives $Q \cdot b(Q-1)/b(Q-1) = Q$. (2) Set $Q = 1$: the numerator is $1 \cdot (0 + 2R) = 2R$ and the denominator is $0 + R = R$, giving 2. (3) Rewrite the mean as

$$\overline{C} = Q + \frac{QR}{b(Q-1) + R}.$$

Since $0 \leq R < b$, the second term is $O(1)$ and hence $\overline{C}/Q \to 1$ as $Q \to \infty$. □

6. Computational Verification

All results have been verified using the `nfield` software engine across multiple bases.

b	p	$p \bmod b$	Q	R	zeros	$\overline{C}$ actual	$\overline{C}$ formula
10	17	7	1	6	9	2.00	2.00
10	61	1	6	0	9	6.00	6.00
10	97	7	9	6	9	9.63	9.63
10	131	1	13	0	9	13.00	13.00
10	179	9	17	8	9	17.81	17.81
10	193	3	19	2	9	19.21	19.21
6	11	5	1	4	5	2.00	2.00
6	41	5	6	4	5	6.71	6.71
6	89	5	14	4	5	14.68	14.68
7	17	3	2	2	6	2.44	2.44
7	41	6	5	5	6	5.76	5.76
7	97	6	13	5	6	13.73	13.73
12	17	5	1	4	11	2.00	2.00
12	67	7	5	6	11	5.56	5.56

Table 1. Gate width and mean formula verified across bases. The zero count equals $b - 1$ in every case, and the mean formula is exact. The third column records the residue class $p \bmod b$.

7. The Two Controls

The autocorrelation of a prime repetend is governed by two independent parameters:

(1) **The base** determines the gate width: exactly $b-1$ lags are blocked, regardless of which prime is being analyzed. This is a property of the coordinate system (the positional representation), not of the prime.
(2) **The prime** determines the gate pattern and intensity. Among the $p-b-1$ constructive multipliers, the distribution of $C(g)$ values carries the structural fingerprint of the prime. The mean intensity satisfies $\overline{C} = Q + O(1)$, so in particular $\overline{C}/Q \to 1$ for large primes, but the distribution around the mean encodes the bin geometry.

This decomposition separates two independent controls. The number of blocked lags is universal $(b-1)$, depending only on the base. The positions of blocked lags depend on p (not just its residue class modulo b), and the mean intensity at constructive lags has an exact closed form in (Q, R).

8. Remarks

The linearization. The proof reveals that the partition geometry, which resisted Fourier, combinatorial, and representation-theoretic approaches, linearizes completely under the change of variables $r \mapsto [br]_p$. In r-space, the bins are contiguous Beatty intervals. In $x = [br]_p$ space, the same bins are the residue classes modulo b. The collision condition $\delta(r) = \delta(gr)$ becomes the congruence $x \equiv [gx]_p \pmod{b}$, and the squeezing argument in Steps 3 and 4 completes the proof.

The explicit deranging set. The deranging multipliers $g = -u/(b-u) \pmod{p}$ for $u = 1, \ldots, b-1$ form a rational family. For $u = b/2$ (when b is even), $g \equiv -1 \pmod{p}$, the complement element. The remaining derangers come in inverse pairs: $-u/(b-u)$ and $-(b-u)/u$ are mutual inverses modulo p.

Generality. The theorem holds for all primes $p > b$, not only those where b is a primitive root. The original computational verification used the primitive-root hypothesis because the autocorrelation $R(\ell)$ is defined on the orbit of b. The algebraic proof bypasses the orbit entirely, working directly with the digit function and the collision count $C(g)$.

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